Photos inside!

When the Earth Moves

Earthquakes, Tsunamis and Volcanoes in Peru

Flyers edition · A2

Before you start

BIG QUESTION

Peru shakes. What do I have to know, and what do I have to have ready, so that it does not catch my family by surprise?

Peru shakes. The ground under the country moves almost every day, and scientists at the Instituto Geofísico del Perú (IGP) write a report for more than two earthquakes a day. Most are small. But five of them changed the country for ever: 1746, 1868, 1970, 2001 and 2007.

This book tells what really happened, using real figures, and explains why it happens. At the end, you will turn everything you know into something useful: a **safety flyer** that a family in your street could follow without asking anybody — even while the ground is moving.

Meet The Aurora Expedition

Ingrid

is the captain and has lots of ideas

Diego

is a reporter and interviews people

Maya

is a scientist with a magnifying glass

Oliver

turns everything into a song

Kili

the condor, brings letters from the Andes

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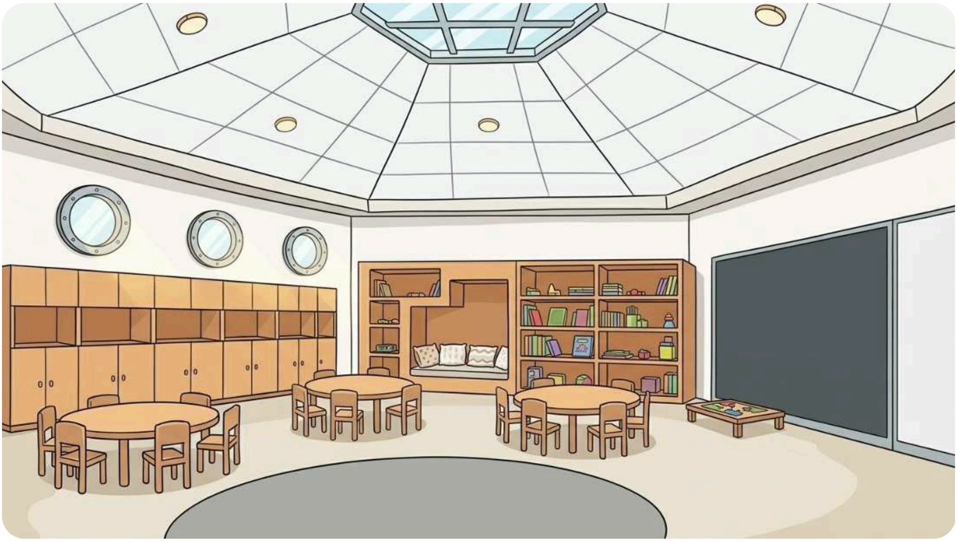
CHAPTER 1

W1 · Lanzamiento

Our Ground Moves



A normal Monday morning at school — until the windows start to rattle.

Why does Peru shake so much? Let's find out!

It was an ordinary Monday morning at a school in Lima. The class was reading in silence when the windows began to rattle. The lamp above the teacher's desk swung gently from side to side, and for a few seconds the floor trembled under everybody's feet.

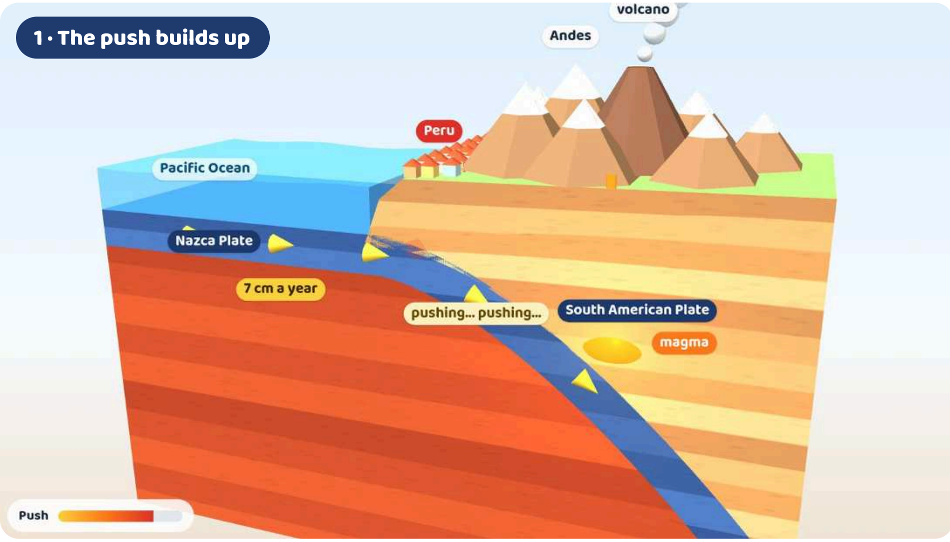
“Is that a lorry going past?” whispered Oliver.

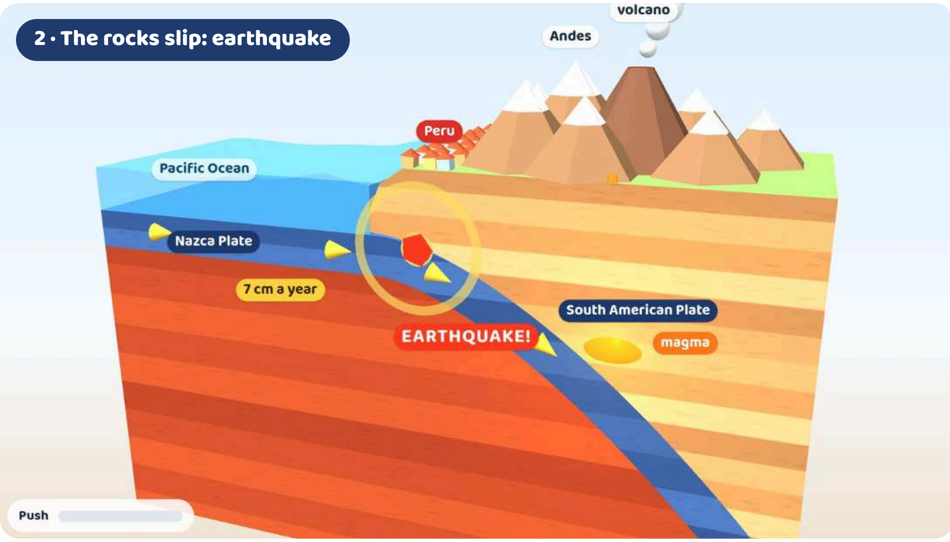
It wasn't a lorry. “It's a *temblor*, a small earthquake,” said the teacher calmly. A few seconds later, everything was still again. Nobody was hurt, and the lesson went on. That afternoon the IGP published a short report about it on its website. For the people of Peru, a day like this is completely normal.

A giant puzzle

The outside of the Earth is not one solid shell. It is cracked into about fifteen giant slabs of rock called **plates**, which fit together like the pieces of a puzzle. The plates sit on hot, soft rock deep inside the Earth, and they are always moving — very, very slowly.

Peru sits on the **South American Plate**. Under the Pacific Ocean, just off the coast, lies the **Nazca Plate**.



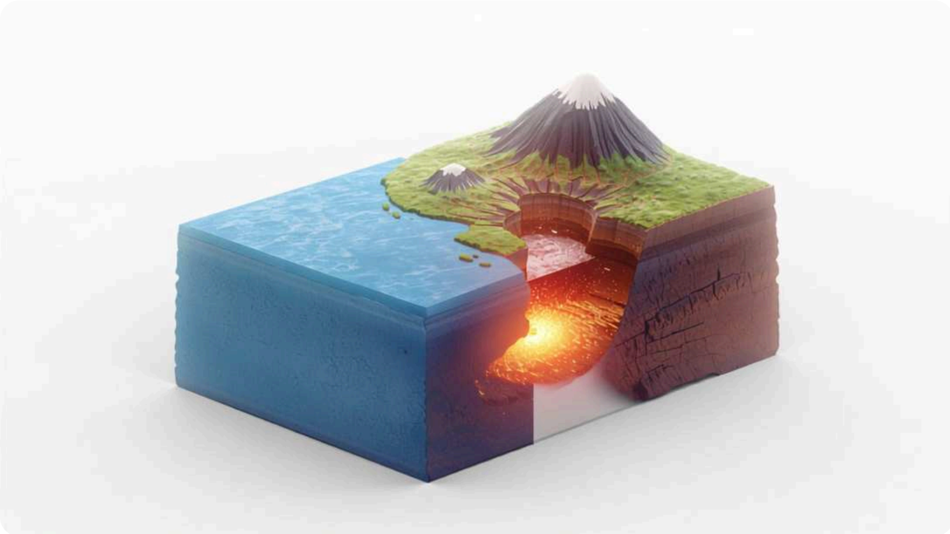


Watch the push build up (the coast is dragged down), then the rocks slip, the land springs back and seismic waves spread out: an earthquake.

3D model — watch it move in the web edition.

The two plates push against each other, and the heavier Nazca Plate slides down underneath South America. Scientists measure this movement with GPS: about 6 to 8 centimetres every year — roughly the length of your finger.

The plates do not slide smoothly, though. Their edges are rough, so they get stuck. For years, even centuries, the push builds up and the rocks slowly bend, like a ruler you press with your thumb. Then, suddenly, the rocks slip. All that stored energy shoots out through the ground in waves, and the ground shakes. That is an **earthquake**. The same push, over millions of years, lifted the land and built the **Andes**, the mountains that run the whole length of Peru.



A slice of the Earth: the ocean plate goes down under the land.

The Ring of Fire

Peru lies on the **Ring of Fire**, a horseshoe of coasts and islands about 40,000 kilometres long that goes around the Pacific Ocean — from Chile and Peru up to Alaska, across to Japan and down to New Zealand. All around it, ocean plates are sliding under other plates. That is why about 9 out of every 10 earthquakes in the world happen on the Ring of Fire, and why it has about 3 out of every 4 of the world's active volcanoes.

The Ring of Fire: about 9 out of 10 earthquakes and 3 out of 4 active volcanoes are on this horseshoe around the Pacific.

3D model — interactive in the web edition.

The Puzzle Man

Photo: Unknown photographer · Public domain

In 1912, a German scientist named Alfred Wegener noticed something strange on a map of the world: the coast of South America and the coast of Africa fit together like two pieces of a puzzle. He also found the same fossils and the same kinds of rock on both sides of the Atlantic. His idea was that millions of years ago all the land formed one giant continent, which he called **Pangaea** ("all lands"), and that the continents have been drifting apart ever since. Most scientists laughed at him. Wegener died on an expedition in Greenland in 1930, before anybody believed him. In the 1960s, scientists studying the ocean floor finally found the proof: the plates really do move.

Two every day

The scientists of the **Instituto Geofísico del Perú (IGP)** watch the ground 24 hours a day. In 2025 they published a report for about 840 earthquakes — an average of more than two a day — and their machines picked up many thousands of smaller ones that nobody could feel. Most quakes are small, or happen far out under the sea or deep underground. But a few, only a handful every century, are strong enough to change the history of the country.

FACT FILE

Nobody in the world — not even the IGP — can say exactly when an earthquake will come. That is why every family must be ready.

The words for what the Earth does

🌍
earthquake
A sudden shaking of the ground when plates slip.

🌊
tsunami
A series of huge waves, usually caused by an earthquake under the sea.

⛰️
landslide
When rocks, soil and mud suddenly slide down a slope. In Peru, a fast river of mud and rocks is called a *huaico*.

🏠
flood
When water covers land that is normally dry — after heavy rain, a tsunami or a river bursting its banks.

Animals and Earthquakes

More than 2,000 years ago, the ancient Greeks wrote that animals sometimes behave strangely before an earthquake. People in Peru tell similar stories: dogs howling, chickens refusing to lay eggs, guinea pigs hiding. Scientists have studied these stories for many years, but so far there is no scientific proof that animals can warn us. The only safe plan is to be ready all the time.

Now you try!

A

Your first page: the four words, each with a drawing and a sentence.

✋ Hand in · W1

Draw each word. Under each drawing, write one sentence that explains it to somebody younger than you.

earthquake	tsunami
landslide	flood

B Match the words and the meanings.

1. plate	a. a line of coasts around the Pacific
2. Ring of Fire	b. a giant piece of the Earth's outside
3. IGP	c. the mountains of Peru
4. Andes	d. the scientists who watch the ground in Peru
5. Nazca Plate	e. the plate under the Pacific next to Peru

C True or false? Correct the false ones.

1. The Nazca Plate slides under the South American Plate. **T / F**
2. Plates move smoothly, so the ground never shakes. **T / F**
3. About 9 out of 10 earthquakes happen on the Ring of Fire. **T / F**
4. Scientists believed Wegener immediately. **T / F**
5. Scientists have proved that animals can predict earthquakes. **T / F**

D Label the diagram of the plates.

Use the 3D picture of the plates. Write: **Nazca Plate · South American Plate · Pacific Ocean · Andes · earthquake**

1 _____________ 2 _____________ 3 _____________

4 _____________ 5 _____________

E Have you felt one? Ask five friends.

Interview five classmates like a reporter: **Have you ever felt an earthquake? Where were you? What did you feel? What did you do?** Then write two sentences about your results: *Most of my classmates… Nobody…*

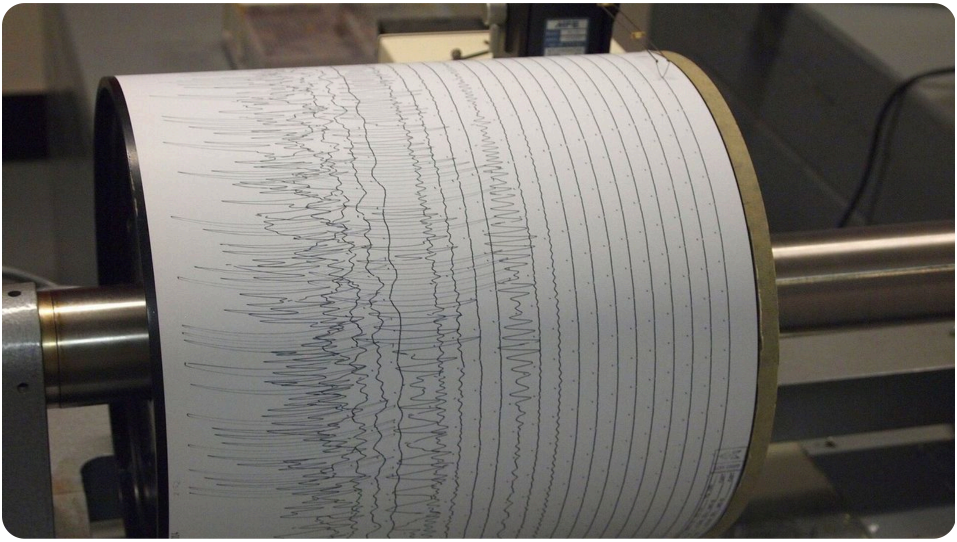
F Maths: how many a week?

In 2025 the IGP reported 840 earthquakes. About how many is that per week? (1 year = 52 weeks) ____ Per day? (365 days) ____ Show how you worked it out.

CHAPTER 2

W2 · Investigación

Reading the Numbers



A seismograph draws the shaking of the ground as a zig-zag line.

Numbers tell the story. Let's interview the data!

Day and night, a network of instruments across Peru is listening to the ground. They are called **seismographs**, and they feel movements far too small for any person to notice. When the ground shakes, the seismograph draws — today on a computer screen, in the past with a pen on a paper drum — a zig-zag line called a seismogram. From these lines, the scientists at the IGP can work out in a few minutes where an earthquake started, how deep it was and how much energy it released. That last number is called the **magnitude**.

Richter Scale


Photo: Gil Cooper, Los Angeles Times · CC BY 4.0

Until 1935 there was no way to compare the strength of two earthquakes with a number. That year, Charles F. Richter (1900–1985), working in California with Beno Gutenberg, invented a magnitude scale. After zero, each whole number means waves ten times taller on a seismograph and about 32 times more energy. So a magnitude 8 earthquake releases about 32 times more energy than a magnitude 7, and about 1,000 times more than a magnitude 6! Today seismologists, including the IGP, use the *moment magnitude* scale, which works better for very big quakes — but many people still call it "the Richter scale".

Magnitude or Intensity?

An earthquake starts on a **fault**, a long crack where two blocks of rock suddenly slip. The **epicentre** is the point on the surface right above that start. The **magnitude** is the energy released, so it is one number for the whole earthquake: Pisco 2007 was 8.0 in Pisco and 8.0 in Lima. The **intensity** is how strongly the shaking was felt in one place, and it changes from one street to the next. After a big earthquake, smaller **aftershocks** can continue for days or weeks. In Spanish people say *sismo*, *temblor* or *terremoto*, but the IGP says they are one thing — and in English there is just one word for all of them: **earthquake**.

Five earthquakes Peru remembers

Of the thousands of earthquakes in Peru's history, these five are the ones the country remembers. A table is the fastest way to compare them.

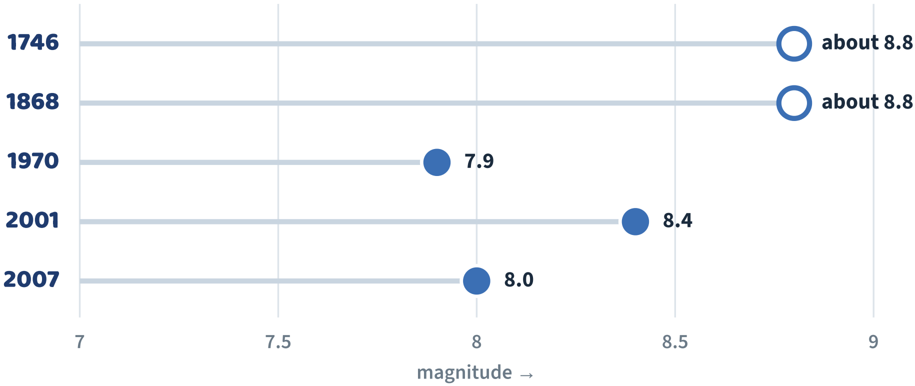
Read along each row to learn about one earthquake; read down each column to compare the same fact for all five.

Five earthquakes Peru remembers

Year	Where	Magnitude	Tsunami?	People who died
1746	Lima and Callao	about 8.8	Yes — about 10 m	about 6,000
1868	Arica (then in Peru)	about 8.8	Yes — about 16 m	thousands
1970	Áncash (Chimbote, Yungay)	7.9	No — an avalanche	about 70,000
2001	Arequipa, Moquegua, Tacna	8.4	Yes — at Camaná	77 (and 68 missing)
2007	Pisco, Ica, Chincha	8.0	Small	596

Magnitudes: USGS (IGP gives 7.9 for 2007). 1746 and 1868 are estimates — there were no seismographs then. Deaths: NOAA, INDECI.

Notice something surprising: the strongest earthquakes are not always the deadliest. The 1746 and 1868 quakes had the highest magnitudes, but the 1970 earthquake — a “smaller” 7.9 — killed far more people than all the others together, because it caused a huge avalanche and because many houses were made of heavy adobe. The magnitude tells you the energy; it does not tell you the whole story. That depends on where the earthquake happens, what time it is, and how the houses are built.



Magnitudes on a scale from 7 to 9. The dots look close, but each whole step means about 32 times more energy. Hollow dots are estimates.

The IGP: Peru's Earthquake Watchers

The IGP began on 1 March 1922, when a small observatory opened in Huancayo, high in the Andes. Today it runs a national network of seismographs and volcano monitors. After an earthquake, the IGP publishes a report within minutes: the time, the place, the depth and the magnitude. Counting its reports year by year gives real data: 798 in 2024 and about 840 in 2025.

FACT FILE

The biggest earthquake ever measured was in Valdivia, Chile, in 1960: magnitude 9.5. Its tsunami crossed the Pacific and reached Hawaii and Japan.

Asking about data

Good researchers ask good questions. Try question words with the table:

When did the Áncash earthquake happen? · **Where** was the tsunami 16 metres high? · **Which** earthquake was stronger, 2001 or 2007? · **How**

many years passed between 1746 and 1868? · **Why** did 1970 have more victims than 1868?
Then compare: *The 2001 earthquake was stronger than the 2007 one, but fewer people died.*

Now you try!

A

Five true sentences taken from the table — one must compare two earthquakes.

✋ Hand in · W2

Write five true sentences using the table. Sentence 5 must compare two earthquakes (*stronger than, more people than, earlier than…*).

B

Read the table and answer.

1. How many years passed between the first and the last earthquake in the table? ________
2. Which earthquake was stronger, 1970 or 2007? ________
3. Which earthquake killed the most people? Why? ______________________
4. Which two earthquakes caused the biggest tsunamis? ________
5. What do the 2001 and 1746 earthquakes have in common? ______________________

C Compare, then explain.

The 2001 earthquake (8.4) was stronger than the 2007 earthquake (8.0), but far fewer people died. Write two or three sentences suggesting reasons. Think about: time of day, where people were, the tsunami, the type of houses.

D Read the chart.

Look at the magnitude chart. The dots look close together, but remember: each step of 1 is about 32 times more energy. Explain in two sentences why a chart of magnitudes can trick our eyes.

E Richter maths

Each whole step means about 32 times more energy. Complete: a magnitude 7 releases about ____ times the energy of a 6; a magnitude 8 about ____ times the energy of a 6 (32 × 32). Round to the nearest hundred.

F Speaking: data detectives

In pairs, choose two earthquakes and prepare a 30-second “news report” comparing them, like Diego. Use: *stronger than, while, but, however*.

CHAPTER 3

W3 · Construcción

What Happened in 1970

Before 1970: the town of Yungay in the valley below Huascarán, the highest mountain in Peru.

This is the worst disaster in Peru's history. Let's tell it carefully, in order.

Sunday, 31 May 1970, began as a bright, sunny day in Áncash, in the north of Peru. In the valley called the Callejón de Huaylas, the town of Yungay lay at the feet of **Huascarán**, the highest mountain in Peru, whose two peaks were covered with ice and snow. It was market day, and the square, with its tall palm trees, was busy. That afternoon the 1970 World Cup was starting in Mexico, so many families had gathered around their radios and televisions. In the town stadium, on a small hill, around 300 children were laughing at a travelling circus.

At 3:23 p.m., about 30 kilometres off the coast near Chimbote, the Nazca Plate suddenly slipped. The earthquake, magnitude 7.9, shook the whole region for about 45 seconds. In Chimbote, Casma and Huaraz, thousands of heavy adobe houses cracked and collapsed, and the air filled with dust.

The shaking broke a giant piece of ice and rock off Huascarán.

Then, high above the valley, the shaking broke off an enormous block of ice and rock from the north peak of Huascarán. As it fell, it picked up snow, mud and boulders as big as houses and turned into a gigantic **avalanche**. It raced down the mountain at around 300 kilometres an hour, jumped over a ridge and, in only about three minutes, travelled roughly 14 kilometres. It buried Yungay and most of the village of Ranrahirca under metres of mud and rock. Only the tops of four palm trees in the square were still visible.

A handful of people saw the grey cloud racing towards them and ran for the cemetery hill, the highest point in Yungay. About 92 people reached the top and survived — among them a young seismologist from the IGP, Mateo Casaverde, who later described what he had seen to the world. About 300 children at the circus in the stadium, also on higher ground, were saved too. But about 20,000 people died in Yungay alone, and in the whole region the earthquake and the avalanche killed around 70,000 people. It remains the worst natural disaster in Peru's history.

The Four Palm Trees

Photo: Yhhue91 · CC BY-SA 4.0

When the mud stopped moving, the only sign of Yungay's main square was the tops of four palm trees sticking out of the ground. Old Yungay was never dug up: it became a *Campo Santo*, a memorial park, where the palm trees still stand today next to the white statue of Christ on the cemetery hill. The people who survived built a new Yungay a short distance away.

Today old Yungay is a memorial park, the Campo Santo, with the statue on the cemetery hill and Huascarán behind.

After the earthquake

Help arrived from every corner of Peru and from dozens of countries: doctors, rescuers, food, water, blankets and tents. The disaster also changed how Peru prepares. In 1972 the government created **INDECI**, the national civil defence institute, and every year on 31 May, the *Día de la Solidaridad* (Day of Solidarity and Reflection on Natural Disasters), people remember the victims and schools practise earthquake drills. Remembering is one way of being ready.

Ingrid's language lab

Past simple — telling it in order

Regular -ed: start**ed**, rush**ed**, bur**ied** (y → ied), travel**led**, destroy**ed**

Say it: /t/ rush**ed** · /d/ bur**ied** · /ɪd/ start**ed**

Irregular: fall → fell · run → ran · break → broke · build → built

Order words: **First**… **Then**… **After that**… **In the end**…

Now you try!

A Your four-sentence report about 1970 — in order, in the past.

✋ Hand in · W3

Write four sentences, in order, in the past simple. Include at least one number and one -ed verb in each sentence. For AD: add a fifth sentence saying what happened afterwards.

B Put the events in order (1–6).

☐ The avalanche covered Yungay.

☐ Families watched the World Cup on TV.

☐ The earthquake started under the sea near Chimbote.

☐ A piece of ice and rock broke off Huascarán.

☐ Peru created INDECI.

☐ About 92 people ran up the cemetery hill.

C -ed sounds: /t/, /d/ or /ɪd/?

Put the verbs in three groups: **started · rushed · buried · helped · travelled · destroyed · shouted · reached · covered · wanted**

/t/	/d/	/ɪd/

D Then and now

Compare the 1970 photo of the statue with today's photo. Write a short paragraph (3–4 sentences) about what changed and what stayed the same.

E A message of solidarity

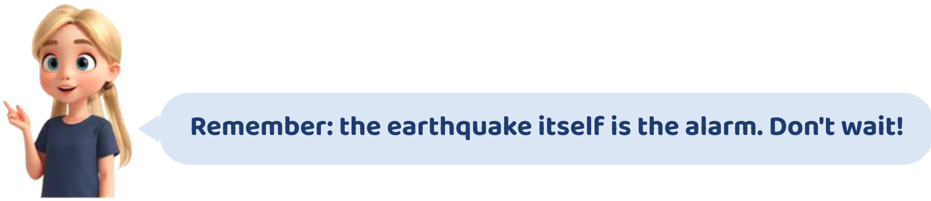
Imagine you are writing a postcard that Kili will carry to Yungay on 31 May. Write 3–4 sentences: what you learned, and why remembering helps us to be ready.

CHAPTER 4

W4 · Borrador

When the Sea Comes

In 1746 a tsunami destroyed the port of Callao.



At about half past ten on the night of 28 October 1746, one of the strongest earthquakes in Peru's history — probably around magnitude 8.8 — shook Lima. Churches, walls and houses collapsed. Around thirty minutes later, the sea rose. A wave about ten metres high swept over the port of Callao, wrecked all the ships in the harbour and carried some of them far inland. Of the five or six thousand people who lived in Callao, only about 200 survived, many by holding on to pieces of wood. After the disaster, the Viceroy, José Antonio Manso de Velasco, led the rebuilding, and a French scientist, Louis Godin, suggested lower buildings and wider streets so that a city could survive the next earthquake.

What is a tsunami?

When the sea floor suddenly jumps up or drops during an earthquake, it pushes the whole column of water above it. This starts a series of waves called a **tsunami**. In the deep ocean the waves travel as fast as a jet plane — around 700 kilometres an hour — but they are low and ships hardly notice them. As they reach shallow water near the coast, they slow down and grow much taller. A tsunami is not a single wave: several waves can arrive over many hours, and the first one is not always the biggest.

> **A Japanese Word**
>
> "Tsunami" comes from Japanese: *tsu* (harbour) + *nami* (wave). Fishermen used the word when they came home to find their harbour destroyed, although they had noticed nothing unusual out at sea. Like Peru, Japan lies on the Ring of Fire.

The ship in the desert

On 13 August 1868, a massive earthquake, probably between magnitude 8.5 and 9.0, struck Arica, which then belonged to Peru. Less than an hour later, a tsunami about 16 metres high rolled in. In the harbour, the American warship *USS Wateree* was lifted up and carried about 430 metres inland — and when the water drained away, it was left standing upright on the sand, with almost all its crew alive. The Peruvian ship *América* was thrown on its side on the beach. The tsunami then crossed the entire Pacific: it reached Hawaii about 14 hours later and New Zealand about 15 hours later. The rusty boilers of the Wateree can still be seen on the beach near Arica today.

The tsunami of 1868 left a warship standing on the beach, far from the sea.

The earthquake is the alarm

If the earthquake starts under the sea close to Peru, the first wave may arrive in just 15 to 20 minutes — too fast for an official warning to reach everybody. That is why the most important rule on the coast is this: **if the shaking is so strong that it is hard to stand up, the earthquake itself is the alarm.** Don't wait for sirens or messages. In June 2001, after the magnitude 8.4 earthquake in southern Peru, the tsunami reached Camaná about 15–20 minutes later and flooded the land for more than a kilometre. Because it was winter the beaches were nearly empty, but even so, about 25 people died. Sometimes the sea pulls back first and leaves the sea floor dry: this is a warning too. Never go to look.

The five steps that save you

1. 🌍 Feel a strong earthquake near the coast? It is your alarm — don't wait for a siren.
2. 🚶 Leave the beach at once, on foot, following the green “Ruta de evacuación” signs.
3. ⬆️ Go up — to high ground or a tall, strong building marked as a safe zone.
4. ⏳ Stay there: more waves can arrive for hours, and the first is not always the biggest.
5. ✅ Return only when the authorities (DHN and INDECI) say it is safe.

A Wave from Far Away

On 29 July 2025, a magnitude 8.8 earthquake struck Kamchatka, in Russia, more than 13,000 kilometres away. Tsunamis can cross oceans, so Peru's Navy (DHN) issued a tsunami alert for the whole coast and 65 ports were closed. The waves reached Callao the next day, about 0.7 metres high. There was no damage — the system worked.

Photo: Gaboct · CC BY-SA 4.0

A

The five steps in order, in your notebook — with a reason for each.

✋ Hand in · W4

Write the five steps in order. After each one, add a reason with *because* or *so that*.

B

Tsunami: true or false?

1. A tsunami is only one wave. **T / F**
2. After an earthquake near Peru, the first wave can come in 15–20 minutes. **T / F**
3. If the sea goes back, it is a good time to look for shells. **T / F**
4. The 1868 tsunami reached New Zealand. **T / F**
5. You should drive your car to the beach to see the waves. **T / F**

C

Read the signs.

Signs must work in seconds. Look at the evacuation sign in the photo and write three things that make it easy to understand quickly (colour, symbols, words, arrow).

D The ship in the desert — questions

1. Why were people surprised when the water went back? ______
2. How long did the tsunami take to reach New Zealand? ______
3. What can you still see near Arica today? ______

E Role-play: on the beach

You are a lifeguard in Chorrillos. A strong earthquake has just happened. In pairs, act out what you say to the people on the beach. Use imperatives and *because*.

CHAPTER 5

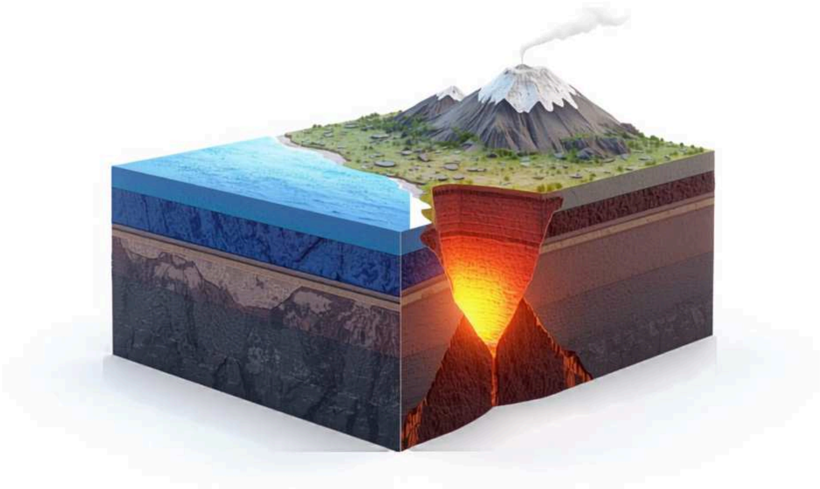
Science link · Reading plan

Mountains of Fire

El Misti (5,822 m) stands above Arequipa, a city of more than a million people.

The same plate that makes earthquakes also makes volcanoes.

A **volcano** is an opening in the Earth's surface where melted rock, gas and ash come out. Underground, the melted rock is called **magma**; once it pours out, it is called **lava**. Many volcanoes grow into tall cone-shaped mountains, built layer by layer from the lava and ash of thousands of eruptions.



Where the ocean plate sinks deep under the land, rock melts and rises as magma.

The Nazca Plate that causes Peru's earthquakes also makes its volcanoes. As the plate sinks about 100 kilometres down, the heat squeezes water out of its rocks. That water helps the hot rock above it to melt, and the magma slowly rises towards the surface. Peru has 16 active or potentially active volcanoes, and the IGP's volcano centre watches 13 of them in real time. Curiously, they are all in the south. Under central and northern Peru the Nazca Plate slides almost flat under the continent instead of diving down steeply, so no magma forms there — the volcanoes of the north went quiet about four million years ago.

The biggest eruption in South America

On 19 February 1600, **Huaynaputina**, a volcano in Moquegua that most people had never noticed, exploded with enormous force. It was the largest eruption in South America's recorded history. Ash and rocks buried about a dozen villages, a thick layer of ash covered Arequipa, and around 1,500 people died. The eruption threw so much ash and gas into the high atmosphere that it dimmed the sunlight around the world: 1601 had one of the coldest summers in centuries in Europe and Asia, and harvests failed in several countries.

An eruption can throw ash high into the sky and cover fields far away.

The White City

Arequipa is known as *la Ciudad Blanca*, the White City, because its cathedral, churches and old houses are built with *sillar*, a light, white stone formed from volcanic ash that hardened long ago. Above it rises El Misti, 5,822 metres high. It has not had a big eruption for more than 500 years, but the IGP watches it closely, because more than a million people live at its feet.

Two volcanoes keep the scientists busy today. **Ubinas**, in Moquegua, is Peru's most active volcano, with about 27 eruptions since the 1550s, the latest between 2023 and 2025. **Sabancaya**, in Arequipa, began erupting on 6 November 2016 and has not stopped: it still puffs small clouds of ash into the sky almost every day. When ash is falling, people near a volcano must stay indoors, close the windows and wear a mask.

FACT FILE

Sabancaya (5,960 m) is higher than El Misti (5,822 m) and Ubinas (5,672 m).

Now you try!

A Label the volcano.

Draw a cross-section of a volcano and label: **crater · ash cloud · lava · magma chamber · sinking plate**. Add one sentence explaining where the magma comes from.

B Superlatives

Write four sentences with superlatives about Peru's volcanoes: *the highest, the most active, the biggest eruption, the most dangerous for a big city*. Explain your last choice.

C True or false?

1. All of Peru's active volcanoes are in the south. **T / F**
2. Huaynaputina erupted in 1970. **T / F**
3. Sabancaya is still erupting. **T / F**
4. Arequipa is called the White City because of snow. **T / F**

D Experiment: make a volcano — and write the rules

Make the volcano with your teacher. Write the method in four imperative steps, then two rules with *must / must not* and one piece of advice with *should*.

When People Get Hurt


A street in Pisco after the earthquake of 15 August 2007. Rescue trucks bring help.

Photo: Senior Airman Shaun Emery, U.S. Air Force · Public domain (U.S. military)

Stay calm, ask 'Where does it hurt?', and tell an adult.

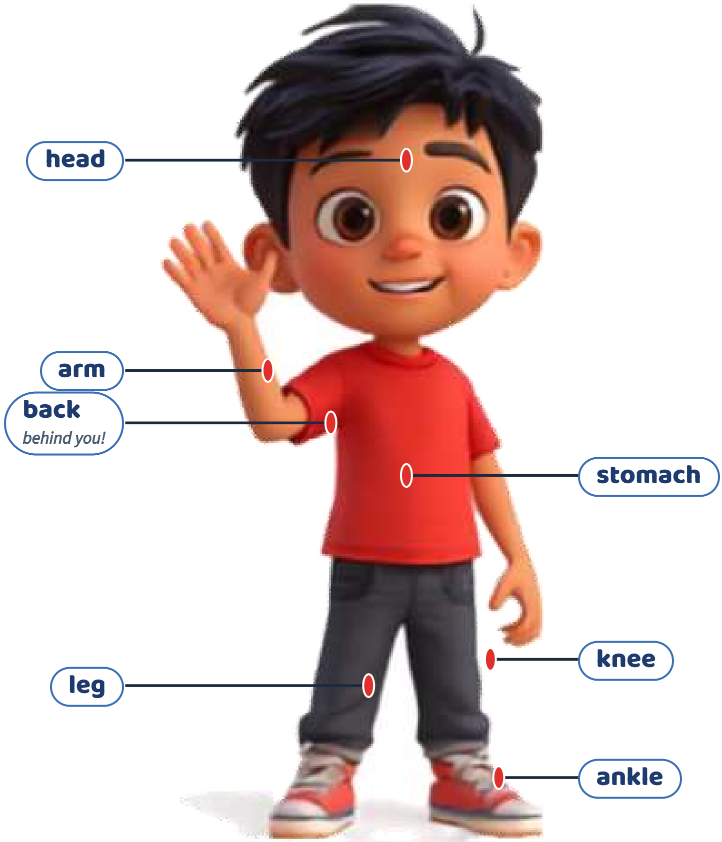
At 6:40 on the evening of 15 August 2007, a magnitude 8.0 earthquake struck just off the coast of Ica. The shaking went on for between two and three and a half minutes — an extremely long time. In Pisco, Chincha and Ica, tens of thousands of houses, many of them built of old adobe, collapsed or became unsafe to live in. In Pisco, the roof and walls of the San Clemente church fell during an evening Mass. In total, 596 people died and 1,292 were injured.

Within hours, firefighters, rescuers with trained dogs, doctors and volunteers were working among the ruins. Planes from Peru and from other countries landed in Lima with food, water, tents and medicine, and long lines of trucks carried them south. The lesson from Pisco is simple: in the first minutes after a disaster, the people who help are usually the people who are already there — neighbours, teachers, classmates. Knowing what to say and whom to tell can make a real difference.

Relief supplies arrive at Lima airport for Pisco, August 2007.

Photo: Tech. Sgt. Jeremy Lock, U.S. Air Force · Public domain (U.S. military)

My body



Parts of the body. The back is behind you, so the line points to where it is.

The first question is always: **Where does it hurt?** Answers use *hurt* or *have got*: *My ankle hurts.* · *I've got a backache.* · *I think I've hurt my knee.* · *I've got a cut on my arm and a bruise on my leg.* A **cut** is when the skin opens and bleeds; a **bruise** is a purple or blue mark under the skin after a knock.

Who do you tell? What do you say?

Tell an adult immediately — a teacher, a member of the school brigade, a parent. Give three pieces of information clearly: **who** is hurt, **where** they are, and **what** happened. Don't try to move somebody who is badly hurt. If an adult needs to call for help, these are the numbers in Peru:

🚒	**116**	Firefighters *Bomberos*
🚓	**105**	Police *Policía*
🚑	**106**	Ambulance *SAMU*
🦺	**115**	Civil defence *Defensa Civil*
📞	**119**	Leave a voice message *Mensajes de voz*

Line 119: Leave a Message

After a large earthquake, phone networks jam as millions of people call at the same time. Peru's free **119** service solves this: you dial 119, choose option 1 and record a message of up to one minute; your family dials 119, chooses option 2, types your number and hears you say you are safe. Agree on this plan with your family before you need it.

Maya: Are you all right? Where does it hurt?

Diego: I've hurt my ankle and I've got a cut on my hand.

Maya: Don't try to stand up. Stay here — I'm going to tell the brigade teacher.

Diego: OK. Thanks for staying calm.

Now you try!

A A four-line dialogue: somebody is hurt and you help.

✋ Hand in · W5

Write a four-line dialogue. Use *Where does it hurt?*, one body word, one ache or injury word, and tell an adult who, where and what.

B Label the body.

Label the picture (**head, arm, leg, back, knee, ankle**), then write where you get a headache, a backache and a stomachache.

C What's the problem?

Match the problem and the help.

1. I've got a cut.	a. Sit down and put something cold on it.
2. I've got a bruise.	b. Wash it and put a plaster on it.
3. My ankle hurts a lot.	c. Don't walk. Tell an adult.
4. I've got a headache.	d. Sit in the shade and drink water.

D Emergency numbers quiz

Complete the table of Peruvian numbers (116, 105, 106, 115, 119) and write one situation for each.

E Who, where, what?

Rewrite this unclear message so a teacher can act in seconds: “Miss! Miss! Come! Something happened, there's a boy, he's crying, I don't know!”

The Backpack

The official INDECI list — for two adults, 24 hours

- water (4 × ½ litre) — *agua embotellada*
- torch and batteries — *linterna y pilas*
- whistle — *silbato*
- tinned food — *comida enlatada*
- masks — *mascarillas*
- toilet paper — *papel higiénico*
- work gloves — *guantes*
- plastic bags — *bolsas*
- first-aid kit — *botiquín*
- small radio — *radio portátil*
- blanket — *manta polar*
- biscuits and chocolate — *galletas y chocolate*
- hand gel — *gel antibacterial*
- emergency phone numbers — *agenda con teléfonos*
- rope (7 m) — *cuerda*
- pencils and a thick pen — *útiles para escribir*

The rule is simple: 24 hours, 8 kilos, one for every two people.

Imagine the first night after a big earthquake. The electricity is off, the taps are dry, the phones are busy and the shops are closed. Your family may have to spend the night outside, far from the kitchen. This is exactly what the **emergency backpack** is for. INDECI, Peru's civil defence institute, publishes an official list so that every family can prepare one before it is needed.

The real INDECI rules

24 h

It must cover the first 24 hours after the emergency.

8 kg

It must not weigh more than 8 kilos, so that you can carry it and run.

1 : 2

One backpack for every two adults.

1 L

About one litre of water per person — four half-litre bottles for two adults.

🎒 The official INDECI list — for two adults, 24 hours

- 💧 water (4 × ½ litre) — *agua embotellada*
- 🔦 torch and batteries — *linterna y pilas*
- whistle — *silbato*
- 🥫 tinned food — *comida enlatada*
- 😷 masks — *mascarillas*
- 🧻 toilet paper — *papel higiénico*
- 🧤 work gloves — *guantes*
- 🛍️ plastic bags — *bolsas*
- 🩹 first-aid kit — *botiquín*
- 📻 small radio — *radio portátil*
- blanket — *manta polar*
- 🍪 biscuits and chocolate — *galletas y chocolate*
- 🧴 hand gel — *gel antibacterial*
- 📒 emergency phone numbers — *agenda con teléfonos*
- 🪢 rope (7 m) — *cuerda*
- ✏️ pencils and a thick pen — *útiles para escribir*

Notice the difference between a **rule** and **advice**:

Rules (must / must not): You must pack water. You must keep the backpack where you can grab it quickly. You must not make it heavier than 8 kilos.

Advice (should): You should check the dates on the food and water twice a year. You should add things for babies, grandparents or pets if your family needs them.

Why a Whistle?

Rescuers searching after an earthquake listen for sounds. Shouting uses a lot of energy and your voice soon gets weak, but a whistle is loud, weighs almost nothing and can be heard far away. Three short blasts is a well-known call for help.

The Reserve Box

The backpack is only half of INDECI's “survival combo”. The other half is the **caja de reserva**, a reserve box that stays at home in a dry, protected place. It holds bigger bottles of water, tinned food, milk, warm clothes, a blanket, soap, a toothbrush and spare batteries for days 2 to 4. You don't carry it when you evacuate — it waits for you.

Now you try!

A

Your list of eight things, each with a reason — using must and should.

✋ Hand in · W6

Write eight items with a reason. Use *must* for at least five and *should* for at least two. Check the total weight is under 8 kg (activity C).

B

Rule or advice?

Decide if each sentence is a rule (must/must not) or advice (should). Rewrite it correctly.

1. You should not make it heavier than 8 kg.
2. You must add a colouring book.
3. You must pack water.
4. You should check the food dates twice a year.

C Maths: under 8 kilos?

Water 2 kg · tins and biscuits 1.8 kg · blanket 1.2 kg · torch, radio, batteries 1.1 kg · first-aid kit 0.6 kg · rope, gloves, knife 0.9 kg · plastic sheet, bags, tape 0.7 kg. Total? How much more can you add?

D Mateo's backpack

Diego packed his backpack for a report. Find five problems and write advice: a 5-litre bottle, no whistle, sweets only, a torch with no batteries, a laptop, no copy of the family's phone numbers.

E Take it home

Show the INDECI list to your family. Tick what you already have and circle what is missing. You will use this in Chapter 12.

The Safe Zone and the Drill

Students in Lima practise at the national drill (simulacro).

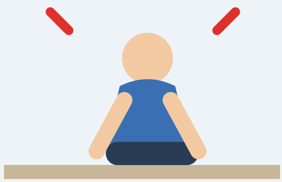
Drop, cover, hold on — I made it into a song!

Look around your classroom: somewhere there is a green sign that says **ZONA SEGURA EN CASO DE SISMOS**. In Peru, green safety signs mean "safe". This one marks the safest places inside a building — beside a strong column, under a beam, and away from windows, glass, tall shelves and anything that could fall. Outside, there is also a **punto de reunión** (meeting point): an open space away from walls, trees and electric cables where everyone gathers after leaving the building.

The green "S" sign marks a safe zone inside a building.

Photo: AgainErick · CC BY-SA 3.0 (also GFDL)

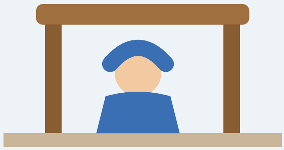
Drop, cover, hold on



DROP

Agáchate

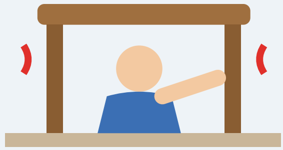
Drop onto your hands and knees before the shaking knocks you down.



COVER

Cúbrete

Cover your head and neck with your arms, or get under a strong desk.



HOLD ON

Sujétate

Hold on to your cover until the shaking stops.

During the shaking, running is dangerous: people fall on stairs and get hit by glass and falling objects near doors. The safest thing is to **drop** onto your hands and knees before the earthquake knocks you down, **cover** your head and neck with your arms (or get under a strong desk), and **hold on** until the shaking stops. In Peru, schools teach it as *agáchate, cúbrete y sujétate*.

The three moments of a Peruvian drill

The three moments of a Peruvian drill

1 🔔 1. PROTECT — When the alarm sounds, drop, cover and hold on in the internal safe zone for as long as the shaking lasts.

2 🚶 2. EVACUATE — When the shaking stops, leave in a line along the marked route: no corro, no grito, no empujo (I don't run, I don't shout, I don't push).

3 🙋 3. MEET AND COUNT — At the meeting point, the teacher and the brigade count everybody and wait for instructions.

Several times a year, the whole of Peru practises at the same moment in the **Simulacro Nacional Multipeligro**. In 2026 there are three: on 29 May, on 14 August and one more later in the year. “Multipeligro” means many dangers: coastal towns add a tsunami evacuation, towns near rivers practise for floods and huaicos, and towns near volcanoes practise for ash. On 29 May 2026, more than nine million schoolchildren took part — which means your drill is part of something enormous.

No corro, no grito, no empujo

The famous Peruvian drill rule — *No corro, no grito, no empujo* — is short so that you can remember it when you are frightened. Each part has a reason: running causes falls on stairs, shouting spreads panic and drowns out instructions, and pushing blocks doors and corridors. A calm line is actually the fastest way to empty a building.

A

The map of your classroom with the safe zone, the way out and the meeting point.

✋ Hand in · W7

Draw an accurate map from above. Include a key (door, windows, columns, safe zone, evacuation route, meeting point) and write two sentences explaining why you chose that safe zone.

B Put the drill in order.

☐ The teacher counts everybody. ☐ The alarm sounds. ☐ We walk out in a line. ☐ We drop, cover and hold on. ☐ The shaking stops. ☐ We wait for instructions.

C Game: Simon says

Play in groups. The leader gives drill instructions (*Drop! Cover your head! Walk to the door!*). Only follow them after “Simon says”. The last player out becomes the leader.

D Oliver's chant — write a verse

Learn Oliver's chant, then write a second verse about the tsunami steps: *When the ground begins to shake, / drop, cover, hold on, for goodness' sake! / No running, no shouting, no pushing, no fear, / walk to the meeting point — we're all here!*

E Before, during or after?

Sort these actions into three columns: *pack the backpack · drop, cover, hold on · go to the meeting point · learn the 119 number · stay away from windows · check if anyone is hurt · agree a family meeting point · stay away from the beach*

BEFORE	DURING	AFTER

How a Safety Flyer Works

A good sign or flyer works in seconds — like this tsunami evacuation sign in Barranco, Lima.

I carry letters — and flyers. A good flyer has headings, pictures and instructions.

Who will read your flyer? A family in your street who has never thought about earthquakes. They will not read a report; they will glance at your flyer on the fridge door, or read it while the ground is actually moving. That changes everything about how you write it. A good safety flyer is built from a few parts that each do one job: a **title** that asks or tells, **headings** that organise, **instructions** that say exactly what to do, **pictures** that show it, and **captions** that explain the pictures.

Peru shakes. Is your family ready?

BEFORE	DURING	AFTER
Ready by the door	**Drop, cover, hold on**	**Out, up and in touch**
Pack the INDECI backpack (water, torch, whistle, first-aid kit) and keep it by the front door. Agree on a family meeting point.	Don't run. Drop to your knees, cover your head and neck, and hold on until the shaking stops. Keep away from windows.	Leave calmly by the stairs. On the coast, go up to high ground at once — don't wait for a siren. Call 119 to leave a message.
The backpack must not weigh more than 8 kg.	*Most injuries happen to people who run.*	*Come back only when the authorities say it is safe.*

Emergency: 116 · 105 · 106 · Voice messages: 119

Model flyer — 119 words.

The parts of a flyer

🔠 **Title**	Large, at the top; often a question that grabs attention.
🏷️ **Heading**	Organises the flyer into the three moments: before, during, after.
👉 **Instruction**	A short imperative: one action per sentence, with the verb first.
🖼️ **Picture**	Shows the action so that even a young child — or someone who reads slowly — understands.
💬 **Caption**	A short line under a picture that explains it or adds one fact.

The hidden instruction

Many real flyers hide their instructions inside information: “Windows can break during an earthquake.” The hidden instruction is **Stay away from windows**. When you read, find the hidden instruction; when you write, put it first. Compare: “It is *recommended that families keep water available*.” → “*Keep water in your backpack*.” The second one works in two seconds.

Now you try!

A

The three headings of your flyer with one instruction under each — verb first.

✋ Hand in · W8

Write three original headings (not just "Before/During/After") with one instruction each. Example: **BEFORE — Ready by the door**: Keep your backpack next to the front door.

B

Label the model flyer.

Write the name of each part next to the model flyer: **title · heading · instruction · picture · caption**. How many instructions are there in total?

C

Find the hidden instruction.

1. Lifts can stop working during an earthquake. → ____________
2. After a strong quake on the coast, the sea can arrive in 15 minutes. → ____________
3. Phone lines are often busy after a big earthquake. → ____________
4. Heavy objects on high shelves can fall. → ____________

D Good flyer or bad flyer?

Read this flyer text: *“Earthquakes are a natural phenomenon that has happened many times in the history of our country and it is very important that all the families think about them.”* Rewrite it as two instructions for a real family.

Writing My Flyer

Plan, write and draw: your flyer, your words.

How about we plan it first? Three parts, 90–110 words, must, must not and should.

Now you become the author. Your flyer must work for a family who has never thought about earthquakes: **three parts** (before, during, after), each with a heading, at least one clear instruction and a drawing, and **90 to 110 words** in total. Use **must / must not** for rules and **should** for advice — and remember the difference.

BEFORE	DURING	AFTER
Heading (original):	Heading (original):	Heading (original):
Instructions — rule (must) and advice (should):	Instructions — rule (must) and advice (should):	Instructions — rule (must) and advice (should):
Drawing + caption:	Drawing + caption:	Drawing + caption:

Word count: ______ (90–110)

Word bank

Disasters earthquake · tsunami · landslide · flood · volcano · ash

Safety safe zone · drill · exit · meeting point · shelter · danger · Drop, cover, hold on

Backpack emergency kit · water · torch · whistle · blanket · first-aid kit · radio

Grammar must · must not · should · have to · before · during · after

Sentence starters

Before: Every family must... · Choose a meeting point that... · You should check the backpack every...
During: Don't run outside — ... · Protect your head and neck by... · If you are near the sea, ...
After: Leave calmly along... · Check that nobody is hurt, then... · Don't come back until...

FACT FILE

Count your words: titles and headings count too. Aim for 90–110.

Now you try!

A Your first draft, with its three drawings.

✋ Hand in · W9

Write your first draft (90–110 words) and draw one picture per part. Underline every *must*, *must not* and *should*. Count your words: ______

B Rule or advice? Explain.

Explain to a partner why *You must keep water in the backpack* is a rule but *You should practise at home* is advice. Write your explanation in two sentences.

CHAPTER 11

W10 · Producto y audiencia

Making It Better

Peer check: a friend reads your flyer like the family in your street would.

Let's check it like editors: parts, verbs, words, voice.

Professional writers never publish a first draft. Swap flyers with a partner and read theirs as if you were the family in the street. Use the checklist, then give feedback as **two stars and a wish** — two specific strengths and one specific improvement.

✅ Checklist

☐ My flyer has a BEFORE part.

☐ My flyer has a DURING part.

☐ My flyer has an AFTER part.

☐ I use must, must not or should.

☐ I say what the family must have ready.

☐ I say where to go.

☐ Every instruction starts with a verb.

☐ It has 90–110 words.

☐ Each part has a drawing.

Check the verbs

Check every instruction: does it begin with a verb (*Pack*, *Keep*, *Drop*, *Walk*, *Don't…*)? Is there only one action per sentence? Replace vague verbs (*do*, *have*, *be careful*) with exact ones (*cover*, *hold on*, *walk*).

Diego's flyer — find the mistakes!

- BEFORE: The backpack should not weigh more than 8 kg.
- DURING: Run to the door as fast as you can.
- DURING: Stand in a doorway.
- AFTER: Take the lift to save time.
- On the coast, wait for the siren before you move.
- Go back home after the first wave.

Read it out loud

At the safety fair you will present your flyer to visitors. Rehearse now: read it aloud so it can be heard from the back of the room. Slow down, pause at each heading, stress the verbs, and look up at your audience at the end of each part. Record yourself if you can, and listen back.

Now you try!

A Your finished flyer. ✋ Hand in · W10

Produce the final version of your flyer with all corrections. It must pass every line of the checklist. For AD: test it on someone younger than you and write down what they understood.

B Two stars and a wish

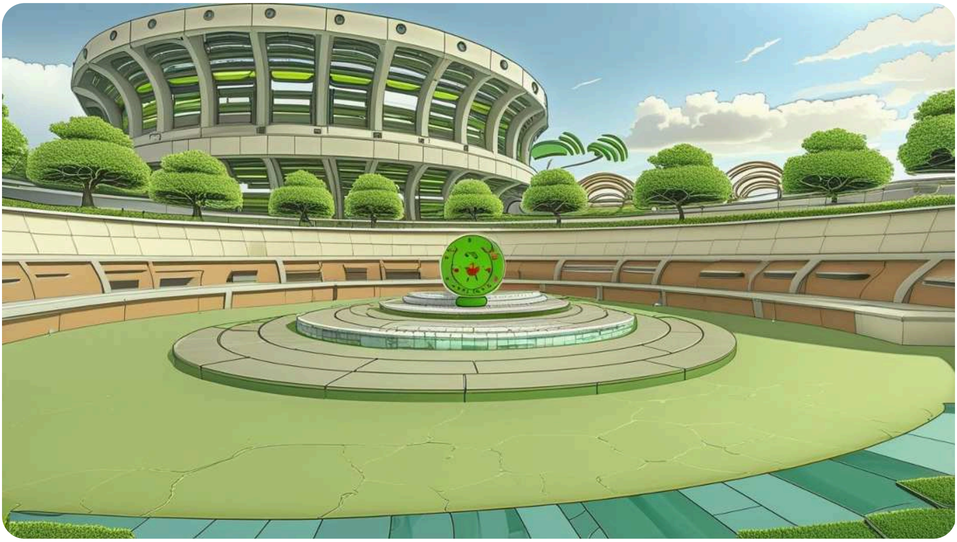
Write feedback for your partner: ⭐ ______ ⭐ ______ 🪄 ______

C Fix the flyer.

Find and correct six mistakes in Diego's flyer (in the box above). Then rewrite the DURING part so it has two clear instructions.

W11 · Producto y audiencia

The Safety Fair



At the safety fair, you present your flyer to people who did not write it.

Big finish! Present it, answer questions, and check your own house.

The safety fair is the real test of your flyer. The visitors — parents, teachers, younger students — did not write it and may never have thought about earthquakes. As they walk around, you will present your flyer in about one minute, answer their questions and ask them what they understood. If your flyer works for them, it will work for a family in your street.

🎤 Presenting my flyer

1. Good morning! My flyer is for a family who has never thought about earthquakes.
2. First, before an earthquake, every family must … .
3. During the shaking, the most important rule is … because … .
4. Afterwards, … — and if you live near the sea, … .
5. What would you do first? Do you have any questions?

And in my house?

The last step is the most personal. Look at your own home with the eyes of a safety inspector. What is already ready? What is missing? And what is the one thing you will actually do about it — not someday, but this week? Three or four honest lines are better than a long list of promises.

At home	We have it ✓	We don't have it ✗
emergency backpack	☐	☐
water	☐	☐
torch and batteries	☐	☐
whistle	☐	☐
first-aid kit	☐	☐
a family meeting point	☐	☐
we know the safe zones at home	☐	☐
we know the 119 number	☐	☐
heavy things are on low shelves	☐	☐

At the start of this book you read a big question: *Peru shakes. What do I have to know, and what do I have to have ready, so that it does not catch my* *family by surprise*? You now know why the ground moves, what happened in 1746, 1868, 1970, 2001 and 2007, what a tsunami is and how to escape it, how to help someone who is hurt, what goes in the backpack and where the safe zone is. Nobody can stop an earthquake. But a family that is ready will not be caught by surprise.

Now you try!

A Your presentation and the family reflection. ✋ Hand in · W11

Present your flyer (about one minute) and answer two visitors' questions. Then write your family reflection in three or four honest lines: what is ready, what is not, and the one thing you will do this week.

B Questions visitors might ask

Write four questions a visitor might ask about your flyer, and prepare an answer to each. Include one difficult question (e.g. “Why not run outside?”).

C How did I do? (Rubric)

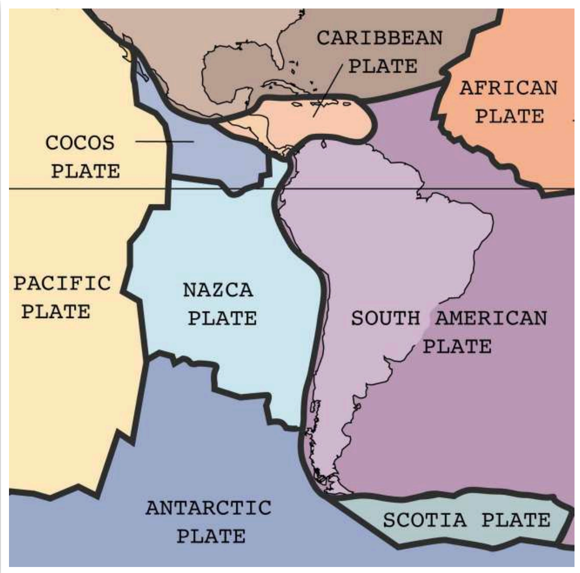
Use the rubric at the back of the book. For each criterion, write AD, A, B or C and one piece of evidence.

Peru in Photos

Real photographs and historical pictures of Peru's earthquakes, tsunamis and volcanoes, from the 1700s to today. Every photo is free to use; the credits say who took it.

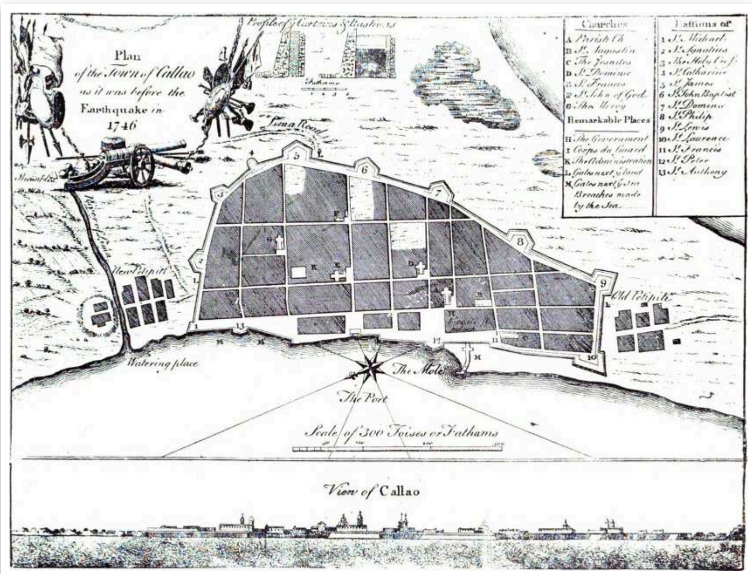
Many old houses in Lima have wooden balconies.

Photo: Felipe Restrpo Acosta · CC BY-SA 4.0



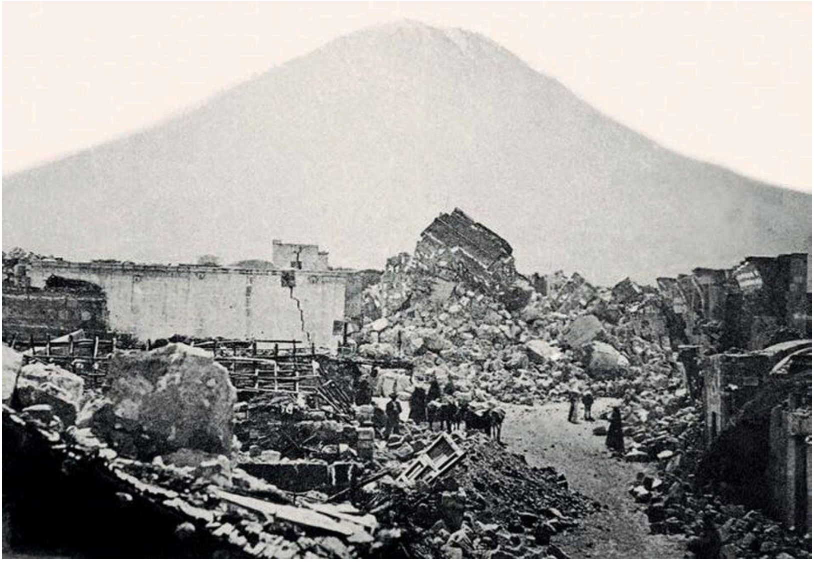
Peru is where the Nazca Plate and the South American Plate meet.

Photo: U.S. Geological Survey · Public domain (USGS)



This old map shows the port of Callao before the earthquake and tsunami of 1746.

Photo: Unknown 18th-century engraver; reproduced by The West Coast Leader, Ltd. · Public domain



The earthquake of 1868 knocked down many buildings in Arequipa.

Photo: Unknown photographer, c. 1868 · Public domain

The tsunami of 1868 carried the ship Wateree onto the land at Arica.

Photo: Unknown photographer, 1868; Henry Clay Cochrane Collection, USMC Archives · CC BY 2.0

Parts of the Wateree are still on the beach near Arica today.

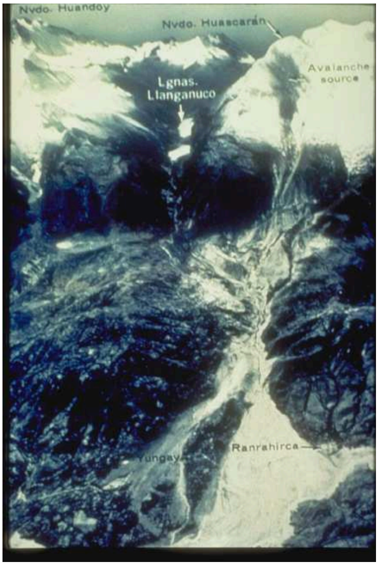
Photo: Pinot · Public domain (released by the author)

The Morro de Arica is a tall rock hill next to the sea.

Photo: Jowyto · CC BY-SA 3.0

Huascarán is the highest mountain in Peru.

Photo: Maurice Chédel · Public domain (released by the author)



From the sky you can see the path of the 1970 avalanche from Huascarán down to Yungay.

Photo: George Plafker, U.S. Geological Survey · Public domain (USGS / NOAA)

After the avalanche of 1970, the statue on the cemetery hill in Yungay was still standing.

Photo: U.S. Geological Survey · Public domain (USGS)

In 1980, people built this gate at the Yungay memorial. Huascarán is behind it.

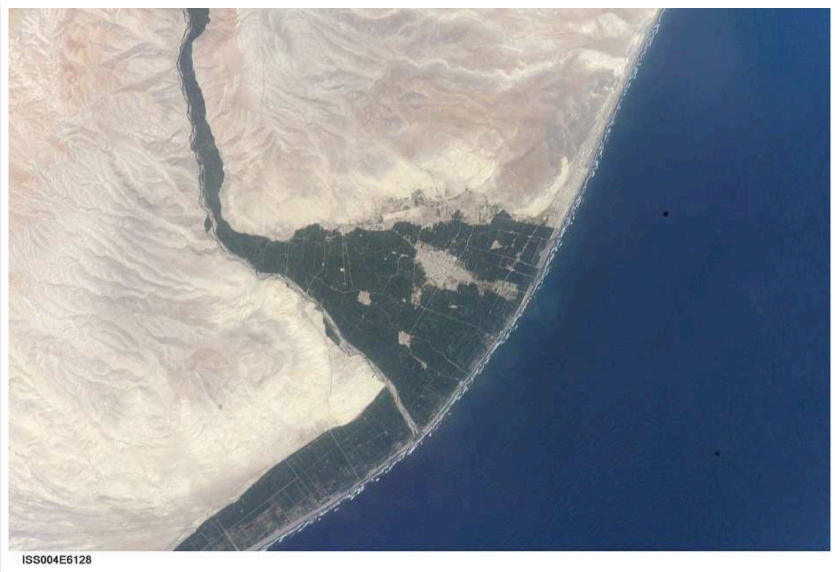
Photo: DB · Public domain (released by the author)

Today the white statue in Yungay looks out at Huascarán.

Photo: Jgaray2000 · CC BY-SA 4.0

These palm trees grew in the main square of old Yungay. They are still there today.

Photo: Yhhue91 · CC BY-SA 4.0



Astronauts took this photo of Camaná, where a tsunami came on land in 2001.

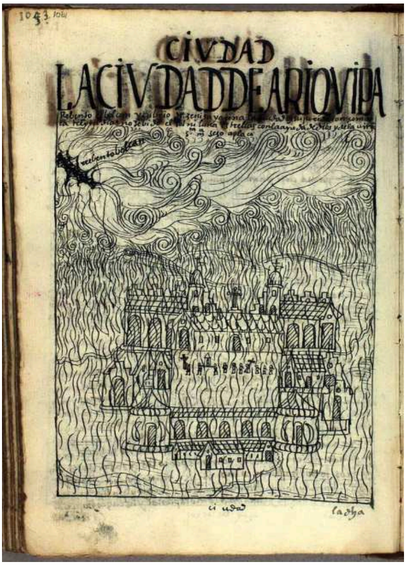
Photo: NASA Earth Sciences and Image Analysis Laboratory, Johnson Space Center · Public domain (NASA)

After the earthquake of 2007, broken bricks filled the streets of Pisco.

Photo: Senior Airman Shaun Emery, U.S. Air Force · Public domain (U.S. military)

Helpers unloaded boxes of supplies for families after the 2007 earthquake.

Photo: Tech. Sgt. Jeremy Lock, U.S. Air Force · Public domain (U.S. military)



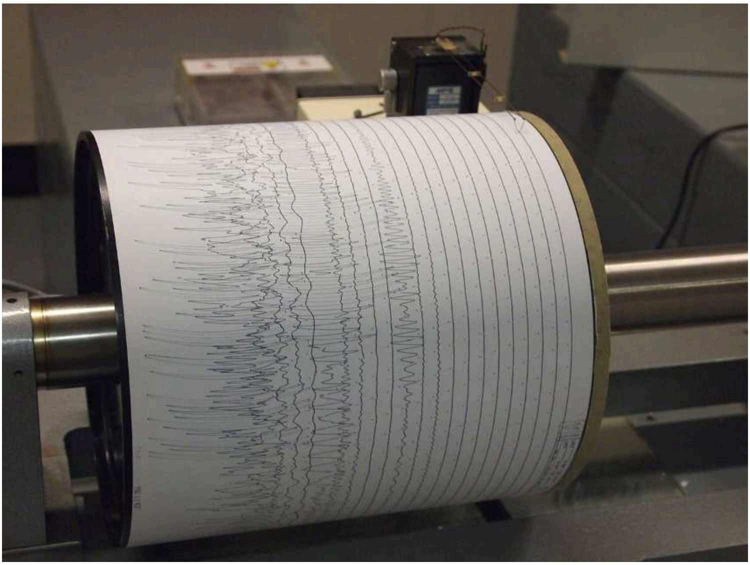
This old drawing shows ash falling on Arequipa when Huaynaputina erupted in 1600.

Photo: Felipe Guaman Poma de Ayala · Public domain

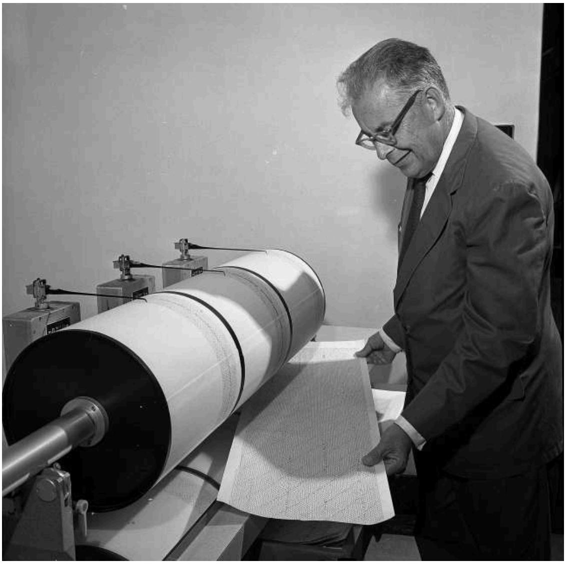
El Misti volcano stands over the city of Arequipa.

Sabancaya volcano sent up a big cloud of ash in 2017.

In 2006, Ubinas volcano blew out a cloud of ash. This photo was taken from space.



A seismograph draws wiggly lines when the ground shakes.



Charles Richter made a scale to tell how strong an earthquake is.

Alfred Wegener said that the continents move very slowly.

In Peru, this sign shows a safe place to go in an earthquake.

This sign in Lima shows which way to go if a tsunami comes.

Students in Lima practise an earthquake drill.

Timeline of Peru's Earthquakes, Tsunamis and Volcanoes

1582 —An earthquake of about magnitude 8 destroys much of Arequipa.

1600 —Huaynaputina erupts — the biggest eruption in South America's recorded history. Ash covers Arequipa.

1604 —An earthquake and tsunami hit Arequipa and Arica; the waves flood about 1,200 km of coast.

1687 —An earthquake and tsunami damage Lima and Callao.

1746 —Lima–Callao earthquake (about 8.8). A 10-metre tsunami destroys Callao; only about 200 people there survive.

1868 —Arica earthquake (about 8.8). A 16-metre tsunami carries the USS Wateree 430 metres inland and reaches Hawaii and New Zealand.

1922 —An observatory opens in Huancayo: the beginning of the Instituto Geofísico del Perú.

1940 —A magnitude 8.2 earthquake shakes Lima; 179 people die.

1970 —Áncash earthquake (7.9). The Huascarán avalanche buries Yungay; about 70,000 people die in the region.

1972 —Peru creates INDECI, its national civil defence institute.

1974 —A magnitude 8.1 earthquake hits Lima and Callao, with a small tsunami at La Punta.

2001 —Southern Peru earthquake (8.4). A tsunami reaches Camaná and a tower of Arequipa cathedral falls.

2007 —Pisco earthquake (8.0): 596 people die and thousands of adobe houses fall.

2016 —Sabancaya volcano begins erupting on 6 November — and is still erupting today.

2019 —A magnitude 8.0 earthquake strikes deep under Loreto, in the Amazon.

2025 —The IGP reports about 840 earthquakes. After a magnitude 8.8 earthquake in Russia, Peru closes 65 ports as a tsunami precaution.

2026 —More than nine million schoolchildren take part in the first national drill of the year, on 29 May.

Timeline of the World

1556 —An earthquake of about magnitude 8 in Shaanxi, China, is the deadliest in history: about 830,000 people die.

1755 —An earthquake of about 8.5, a tsunami and fires destroy Lisbon, Portugal, and lead to scientific studies of earthquakes.

1906 —A magnitude 7.9 earthquake and fires destroy much of San Francisco, USA.

1912 —Alfred Wegener suggests that the continents move.

1935 —Charles Richter and Beno Gutenberg invent a scale to measure the size of earthquakes.

1960 —Valdivia, Chile: magnitude 9.5, the most powerful earthquake ever recorded. Its tsunami reaches Hawaii and Japan.

1964 —A magnitude 9.2 earthquake in Alaska, the biggest in North America.

2004 —The magnitude 9.1 Sumatra earthquake causes the deadliest tsunami ever recorded, around the Indian Ocean.

2010 —A magnitude 8.8 earthquake in Maule, Chile; its tsunami also reaches the coast of Peru.

2011 —A magnitude 9.1 earthquake hits Japan; the tsunami causes a nuclear accident at Fukushima.

2015 —A magnitude 7.8 earthquake in Nepal causes an avalanche at Everest Base Camp.

2023 —Two powerful earthquakes (7.8 and 7.5) hit Türkiye and Syria; about 59,000 people die.

2025 —A magnitude 8.8 earthquake in Kamchatka, Russia, sends tsunami alerts all around the Pacific — including Peru.

Glossary

The words of this unit, with Spanish.

after *(después)* — at a later time: after the earthquake

aftershock *(réplica)* — a smaller earthquake that comes after a big one

ankle *(tobillo)* — the part of the body between your leg and your foot

ash *(ceniza)* — the grey powder that comes out of a volcano

avalanche *(avalancha / aluvión)* — a very large amount of ice, snow and rock falling fast down a mountain

back *(espalda)* — the part of your body behind you, from your neck to your bottom

backache *(dolor de espalda)* — a pain in your back

before *(antes)* — at an earlier time: before the earthquake

blanket *(manta)* — a warm cover for your bed or for sleeping outside

bruise *(moretón)* — a blue or purple mark on your skin after a knock

cut *(corte)* — an opening in your skin that bleeds

danger *(peligro)* — something that can hurt you

drill *(simulacro)* — practice for what to do in an emergency

Drop, cover, hold on *(Agáchate, cúbrete, sujétate)* — what you do during the shaking

during *(durante)* — while something is happening

earthquake *(sismo / temblor / terremoto)* — a sudden shaking of the ground, big or small: in English it is always “earthquake”

emergency kit *(mochila para emergencias)* — a bag with the things you need in an emergency

epicentre *(epicentro)* — the place on the ground right above where an earthquake starts

exit *(salida)* — the way out of a building

fault *(falla)* — a long crack in the rock where the ground can suddenly move

first-aid kit *(botiquín)* — a box with plasters, bandages and things to help people who are hurt

flood *(inundación)* — a lot of water covering land that is usually dry

have to *(tener que)* — it is necessary: we have to leave now

head *(cabeza)* — the top part of your body, with your eyes and mouth

headache *(dolor de cabeza)* — a pain in your head

hurt *(doler / lastimarse)* — to feel pain: my knee hurts

intensity *(intensidad)* — how strongly the shaking is felt in one place

knee *(rodilla)* — the middle of your leg, where it bends

landslide *(deslizamiento / huaico)* — rocks and earth sliding down a hill

lava *(lava)* — hot liquid rock that comes out of a volcano

magnitude *(magnitud)* — a number that shows how strong an earthquake is

meeting point *(punto de reunión)* — the safe place outside where everyone meets after leaving

must / must not *(debe / no debe)* — for rules: you must pack water; you must not run

plate *(placa tectónica)* — a giant piece of the Earth's outside

Ring of Fire *(Cinturón de Fuego)* — the line of coasts around the Pacific with many earthquakes and volcanoes

safe zone *(zona segura)* — the safest place to be in an earthquake

seismograph *(sismógrafo)* — a machine that measures the shaking of the ground

shelter *(refugio / albergue)* — a safe place to stay

should *(debería)* — for advice: you should check your backpack

stomachache *(dolor de estómago)* — a pain in your stomach

torch *(linterna)* — a small light you carry in your hand

tsunami *(tsunami / maremoto)* — very big waves caused by an earthquake under the sea

volcano *(volcán)* — a mountain where hot rock, gas and ash come out

whistle *(silbato)* — a small thing you blow into to make a loud sound

How am I doing?

The unit rubric. AD is the highest level.

I can...	AD	A	B	C
I name natural disasters, safety words, the things in the emergency kit and the parts of the body.	I use them and I explain what one of them means to somebody who does not know it.	I use the words on my own, in sentences, when I am talking about safety.	I say most of the words when I see the picture.	I say a few of the words with help.
I read a table and a chart about earthquakes in Peru and I say what they show.	I notice something in the data that nobody pointed out to me.	I ask and answer questions about the data and I compare two earthquakes.	I find information when I am asked a direct question.	I find one piece of information if somebody shows me the column.
I tell what happened in a real Peruvian earthquake using the past simple.	I tell it in order and I add what happened afterwards.	I tell what happened in four sentences, in order and in the past.	I write two or three sentences in the past with some mistakes.	I copy sentences in the past.
I use must, must not and should to give safety rules and advice.	I explain to somebody else why one is a rule and the other is advice.	I choose must, must not or should correctly and I give the reason.	I write rules but I mix up must and should.	I copy a rule that is given to me.
I organise safety instructions into before, during and after.	I notice when an instruction could go in two moments and I say why.	I sort them all and I use the three moments to organise my own flyer.	I sort most of the actions with a bit of help.	I put two of the three moments in the right place.
I write a safety flyer of 90 to 110 words and I present it to other people.	My flyer works for somebody who is younger than me, and I prove it.	I write the three parts, with a heading, an instruction and a drawing each, and I present it.	I write the three parts but they are short or unclear.	I write some sentences and I show my drawing.

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When the Earth Moves

Peru sits where the Nazca plate dives under South America, so the ground here is never really still. Follow five earthquakes that changed the country, discover how scientists measure them and why tsunamis reach our coast, and find out what INDECI says every family needs. Then turn it all into a safety flyer that really works.

Inside this book

- 12 chapters, one for each week
- Activities after every chapter
- Peru in Photos
- Two timelines
- Glossary with Spanish
- How am I doing? rubric

Flyers edition · A2

Nordic International School · Grade 3 · Unit 5 · Lima, Peru